

Deep Learning Guided Discovery and FEP Validation of NOD2 Binders for the R702W Crohn's Variant

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INTRODUCTION

Background

Over **1 million Americans** suffer from **Crohn's disease**, a painful, incurable inflammatory bowel condition. The **NOD2 gene** is the strongest genetic risk factor. The **R702W mutation** affects **~10% of patients**, increasing disease risk **~2-fold**. **No approved drugs directly target the NOD2 protein itself**.

NOD2 is an innate immune sensor that recognizes bacterial muramyl dipeptide (MDP) and activates **NF-κB signaling** to mount an antimicrobial response essential for intestinal bacterial clearance and immune homeostasis.

Impaired NOD2 causes chronic intestinal inflammation, the defining hallmark of Crohn's disease.

The R702W mutation lies **79.4 Å** from the binding site, suggesting **allosteric mechanisms** may drive its clinical impact. Identifying compounds that maintain binding to R702W could **provide a molecular basis for restoring immune function in mutation carriers**.

Research Question

Can computational methods identify compounds that directly bind and modulate NOD2?

No approved drugs target NOD2 directly. Current therapies (biologics, immunosuppressants, steroids) suppress the immune system downstream, treating symptoms rather than the root cause.

This project explores whether **structure-based virtual screening** can identify NOD2-binding compounds that maintain predicted affinity for **both wild-type and R702W mutant** forms of the protein.

Hypothesis: (1) CNN-based docking + ML can prioritize NOD2 binders
(2) MD simulations will confirm stable binding (3) FEP can quantify mutation effects on affinity

Motivation

As an R702W carrier myself, I wanted to understand why this mutation matters for drug response.

The R702W variant is one of the most common NOD2 mutations in Crohn's patients, yet we don't fully understand how it affects potential therapeutics at the molecular level.

This project aims to bridge that gap, moving toward **precision medicine** that accounts for patient genetics.

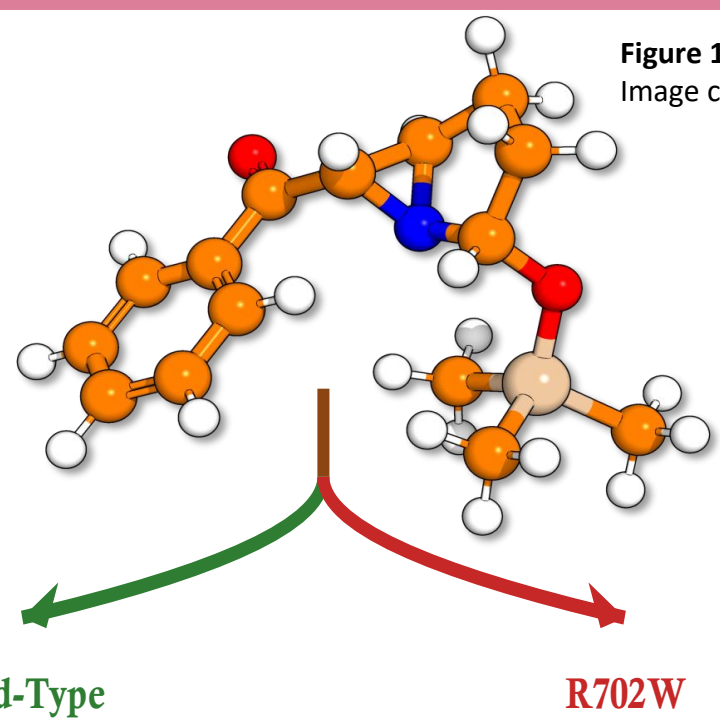
NOD2 Structure and Pathway

MDP
Muramyl Dipeptide: Bacterial Cell Wall Fragment

Smallest immunostimulatory fragment from peptidoglycan
Present in **all bacteria** (Gram+ and Gram-), detected by **NOD2**

→ **Binds NOD2 LRR Domain (res. 744-1040)**
→ **Recruits RIPK2** → **Activates NF-κB**
→ **Cytokine Release (IL-8, TNF-α, IL-1β)**

R702W impairs this pathway → chronic intestinal inflammation



Wild-Type NOD2

DOMAIN ARCHITECTURE
CARD NBD HD1 HD2 LRR

MDP binding triggers conformational change
→ **CARD domain recruits RIPK2**

Wild-Type Pathway

Figure 2: Wild-type NOD2 full-length protein. Image created by finalist using PyMOL, 2026.

Figure 4: Active RIPK2 protein structure. Image created by finalist using PyMOL, 2026.

Figure 6: Active TAK1 protein structure. Image created by finalist using PyMOL, 2026.

Figure 8: Active NF-κB protein structure. Image created by finalist using PyMOL, 2026.

EFFECTIVE IMMUNE RESPONSE
Bacterial clearance + Cytokine production (TNF-α, IL-1β, IL-6) → Intestinal homeostasis maintained

R702W Mutant NOD2

DOMAIN ARCHITECTURE
CARD NBD HD1 HD2 LRR Mut

R702W in HD2 allosterically impairs MDP sensing
→ **Weakened RIPK2 recruitment (~2x Crohn's risk)**

R702W Pathway

Figure 3: R702W mutant NOD2 with mutation site. Image created by finalist using PyMOL, 2026.

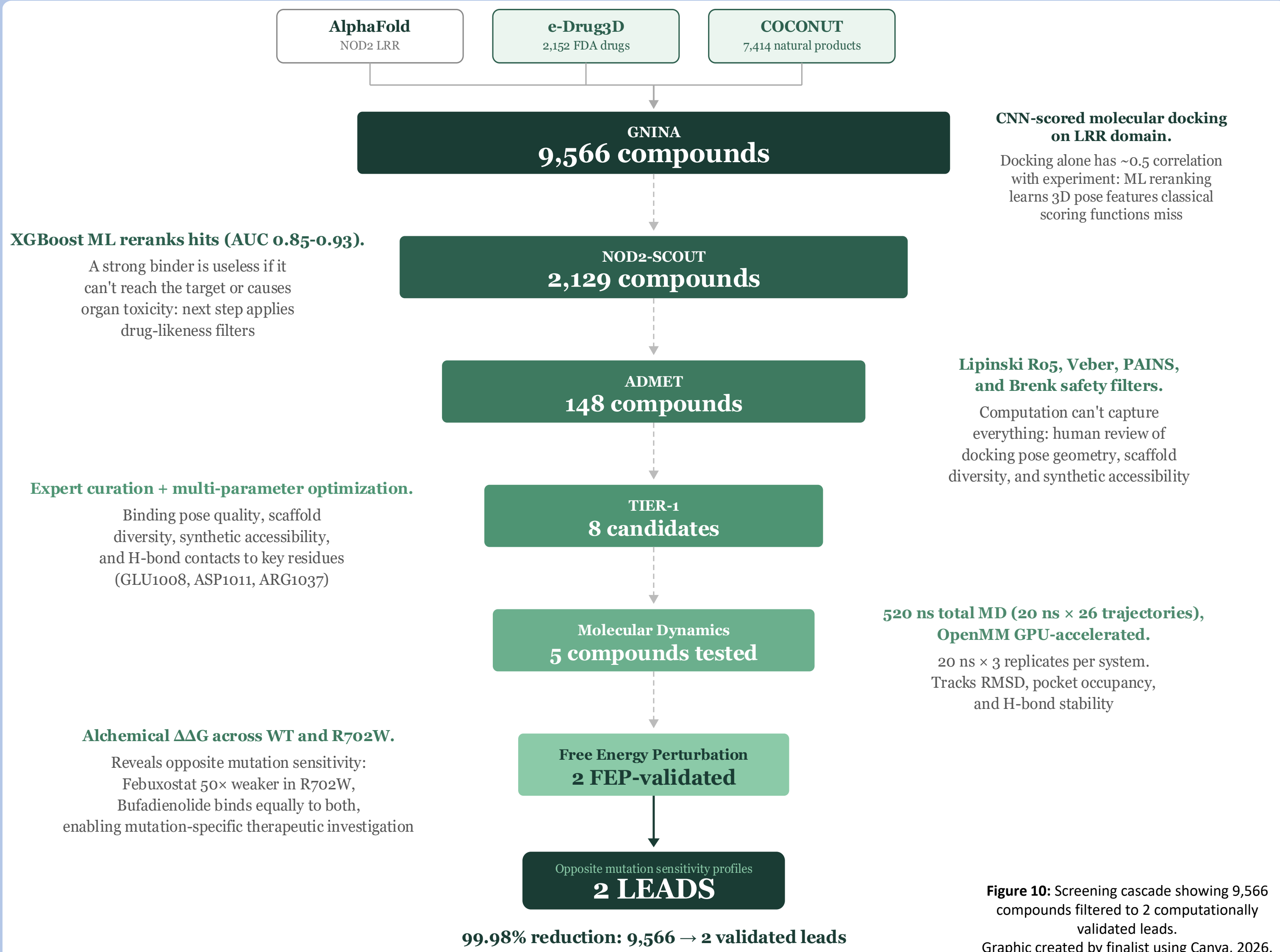
Figure 5: Impaired RIPK2 protein structure. Image created by finalist using PyMOL, 2026.

Figure 7: Impaired TAK1 protein structure. Image created by finalist using PyMOL, 2026.

Figure 9: Impaired NF-κB protein structure. Image created by finalist using PyMOL, 2026.

CRHOHN'S DISEASE
Defective bacterial clearance + Chronic inflammation → Increased disease risk + Intestinal dysbiosis

METHODS



1. PREPARATION

Structure prediction and compound library assembly

No ligand-bound human NOD2 LRR structure → used **AlphaFold2** prediction. Predicted full-length **NOD2 (1,040 residues)** from UniProt Q9HC29. Focus: **LRR domain** (residues 744-1040), pLDDT >90 = high confidence. Prepared with polar hydrogens + Gasteliger-Marsili partial charges.

e-Drug3D: 2,152 FDA-approved drugs (known safety profiles). **COCONUT**: 7,414 natural products (unique scaffolds). 3D structures via **RDKit ETKDG + MMFF94** minimization.

Total: **9,566 unique compounds** for screening.

2. GNINA

Fast docking to rank binders

GNINA: next-gen docking with **CNN scoring** trained on PDBbind. Unlike AutoDock Vina/Glide, CNNs learn binding patterns from real structures.

Grid: **30x30x30 Å** centered on LRR domain (residues 744-1040). **Exhaustiveness = 32** (4x default) for thorough conformational sampling. **9 binding poses** per ligand to capture different orientations.

CNN affinity scores: **2.47–7.34 kcal/mol** (mean: 4.57).

All **9,566 compounds** docked and ranked.

3. NOD2-SCOUT

ML reranks hits (docking alone has ~0.5 correlation with experiment)

XGBoost classifier (100 trees, depth=4, lr=0.05) on docking pseudo-labels: Top 20% = binders, bottom 20% = non-binders, stratified by **5 MW bins**. **856 labeled compounds** (728 training, 128 test).

Features: **2,048-bit Morgan fingerprints + 10 descriptors = 2,058 total**. **5-fold scaffold-split CV**: entire scaffolds held out (stricter than random). Performance: **AUC = 0.85–0.93**. Shuffled control: **AUC = 0.50** (random).

Filtered to top **2,129 compounds**.

4. ADMET

Safety and drug-likeness filter

A strong binder is useless if it can't reach the target or causes toxicity.

Lipinski's Rule of 5: MW ≤500, LogP ≤5, HBD ≤5, HBA ≤10. **Veber's rules**: TPSA ≤140 Å² (oral bioavailability), rotatable bonds ≤10. **PAINS**: removes false-positive-prone compounds (aggregation/redox). **Brenk filters**: flags known toxic substructures (>2 alerts = removed).

All filtering via **RDKit cheminformatics toolkit**.

Result: **148 of 150 passed (98.7%)**.

5. TIER-1

Expert curation (computation can't capture everything)

Manual curation of **148 ADMET-passing compounds** using multi-parameter optimization.

Structural diversity: Tanimoto clustering on Morgan fingerprints. Diverse scaffolds = higher odds of success in experimental validation.

Binding pose quality: H-bonds to **GLU1008, ASP1011, ARG1037**; π-stacking; hydrophobic burial. Strained/exposed poses deprioritized.

Synthetic accessibility: FDA drugs available; natural products vary.

Selected **8 candidates** → **5** advanced to MD.

6. MOLECULAR DYNAMICS

Checks if predicted binding is stable over time (proteins are dynamic)

OpenMM GPU-accelerated (~100 ns/day on RTX). **20 ns × 3 replicates** per system.

Force fields: **AMBER14** (protein), **OpenFF 2.1.0** (ligand), **TIP3P-FB** water. **310 K, 0.15 M NaCl** (physiological). **520 ns total across 26 trajectories**.

Pocket occupancy: % frames ligand stays within **5 Å** of start position. Negative control **Cysteamine** (rank #2056/129): **0% occupancy**. Validates pocket is specific—high occupancy = genuine binding.

Top **2 computationally stable** (70-80% occupancy, ~2 H-bonds).

7. FREE ENERGY PERTURBATION

Gold standard for predicting relative binding affinity differences

FEP "alchemically" transforms ligand → nothing, measuring ΔG in bound vs free state. **120 lambda windows** total (20/leg × 3 legs × 2 compounds) **MBAR** estimator.

ΔΔG = ΔG(R702W) - ΔG(WT): positive = mutation weakens binding. **Febuxostat**: ΔG(WT) = -10.36, ΔG(R702W) = -8.02 kcal/mol. ΔΔG = **+2.34 kcal/mol** (8σ, p<0.001) → **50x weaker predicted binding** to mutant.

Bufadienolide: ΔG(WT) = -15.22, ΔG(R702W) = -15.66 kcal/mol. ΔΔG = **-0.44 kcal/mol** (1.8σ, NS) → binding unaffected.

Key insight: mutation effects are **ligand-dependent!**

8. 2 COMPUTATIONALLY VALIDATED LEADS

Two compounds with opposite mutation sensitivity profiles

99.98% reduction: 9,566 → 2 FEP-validated leads.

LEAD 1: Febuxostat (FDA-approved gout drug)
Mutation-SENSITIVE: 50x weaker to R702W. Mutation is **79.4 Å away!** → Probe for studying allosteric effects of distant mutations.

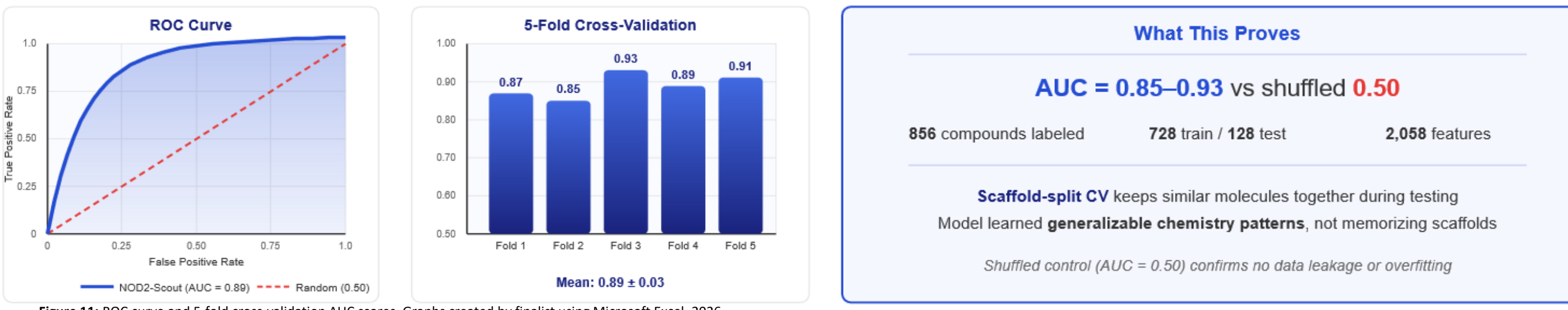
LEAD 2: Bufadienolide (CID 10120, natural cardiac glycoside)
Mutation-INSENSITIVE: binds equally to WT and R702W. → Starting scaffold for mutation-resistant therapeutic development.

Mutation effects are ligand-specific → precision medicine insight.

RESULTS

MACHINE LEARNING VALIDATION

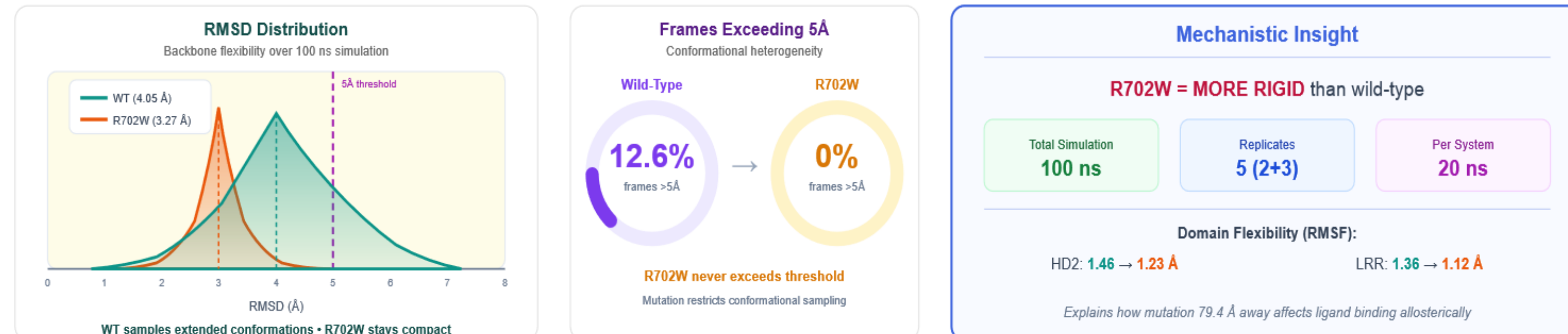
Scaffold-split cross-validation confirms model learned real patterns, not random noise



RESULTS COUNT.

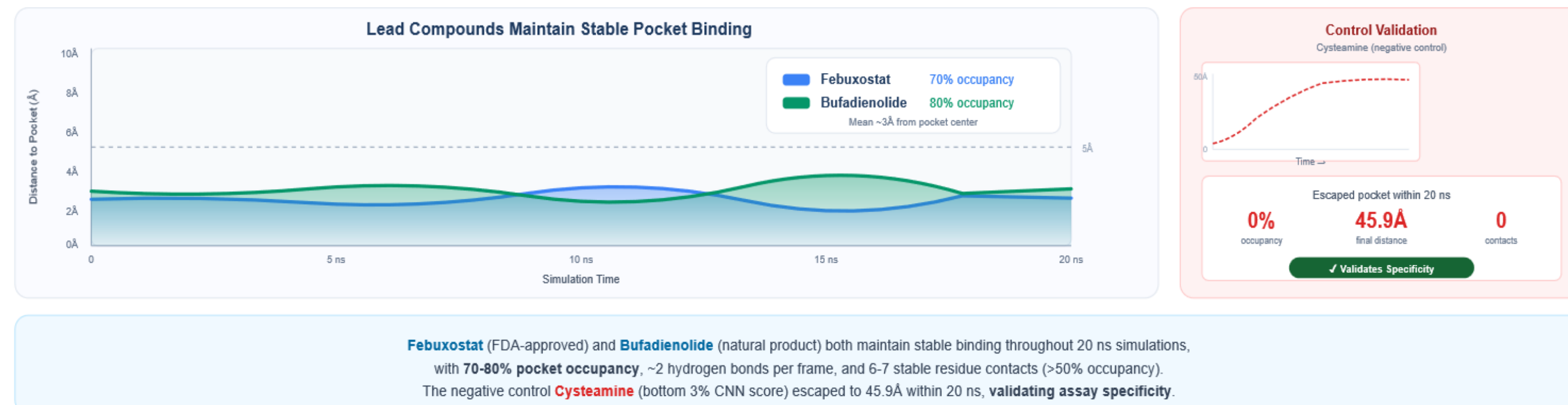
APO DYNAMICS: WT vs R702W

The R702W mutation rigidifies the protein, reducing conformational sampling



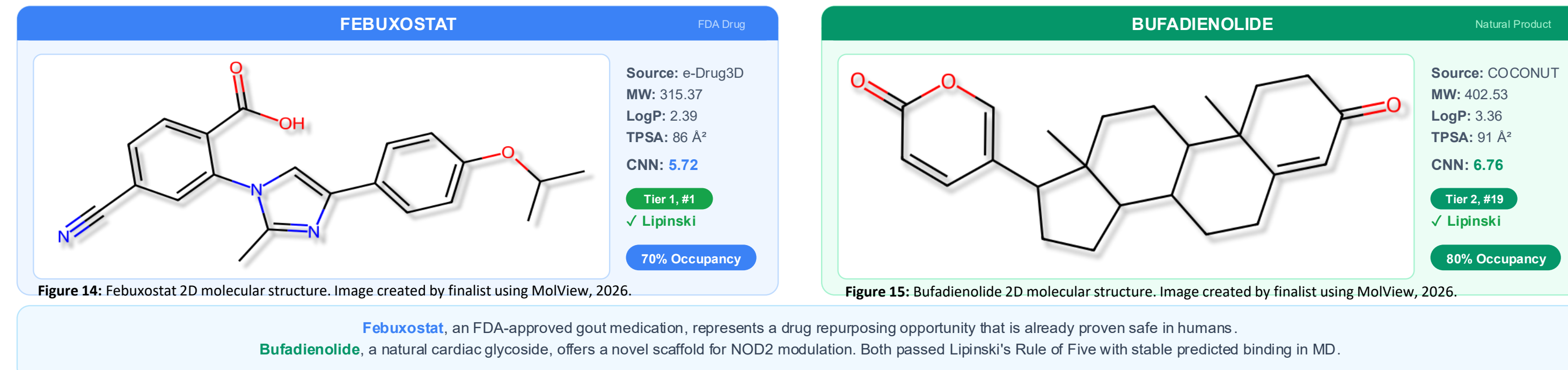
MOLECULAR DYNAMICS VALIDATION

520 ns total simulation across lead compounds and negative control



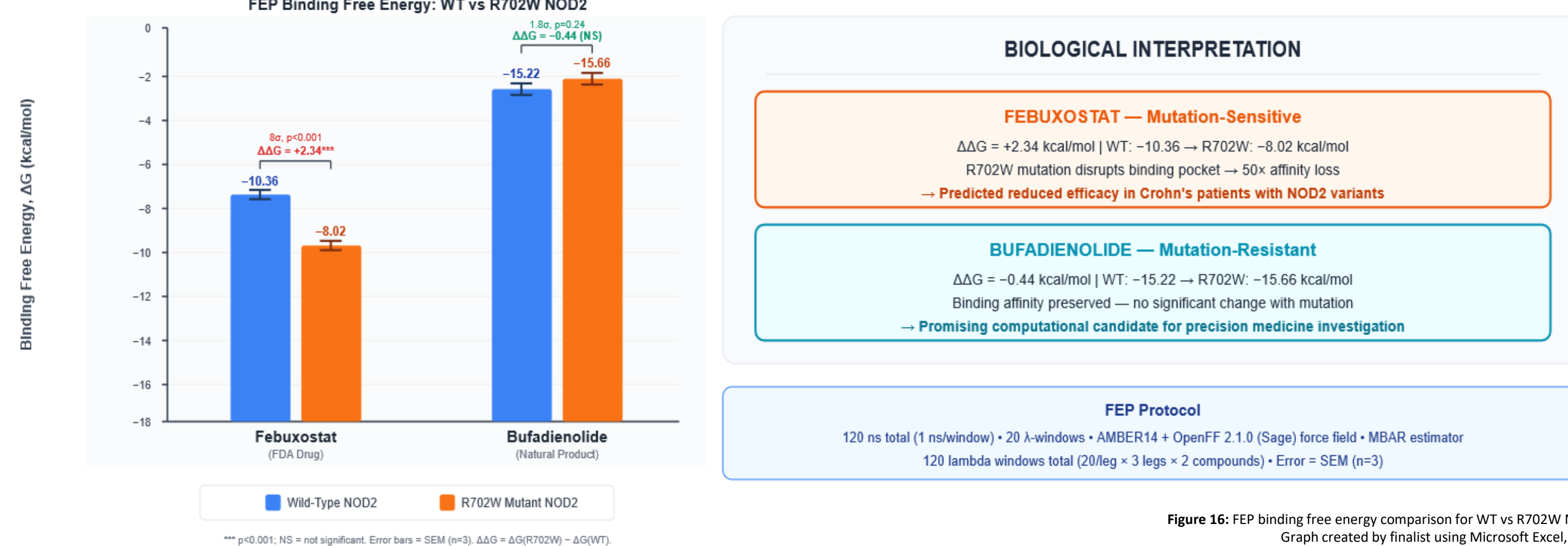
LEAD COMPOUNDS

Two chemically diverse hits from FDA drug library and natural products database



FREE ENERGY PERTURBATION RESULTS

Comparing Predicted Drug Binding Affinity: Wild-Type vs. R702W Mutant NOD2



DISCUSSION

Why does Febuxostat fail at R702W?

R702W substitutes charged arginine (+1) with neutral tryptophan, eliminating 2-3 hydrogen bonds. The lost positive charge disrupts electrostatic networks propagating **79.4 Å** to the LRR binding pocket. FEP: **ΔΔG = +2.34 kcal/mol** (p<0.001) = **50x weaker predicted binding** to R702W mutant vs wild-type NOD2.

Allosteric mechanism: rigidification

APO molecular dynamics simulations suggest R702W **rigidifies** the NOD2 protein: 0% vs 12.6% of trajectory frames exceed 5Å RMSD. Mean RMSD values: 3.27 Å vs 4.05 Å (wild-type is more flexible). Reduced conformational flexibility may impair **induced-fit binding** for ligands requiring protein adaptation to bind optimally. Challenges prior literature that described R702W as "destabilizing" the protein.

Limitations

Computational study requires **SPR binding** and **NF-κB reporter assay** validation. Only R702W tested; G908R and L1007fs remain. No crystal structure available.

Why does Bufadienolide maintain binding?

Completely distinct contact profile from Febuxostat: **ARG1034: 82%** vs 3% occupancy and **ARG1037: 81%** vs 71%. The hydrophobic steroid scaffold relies primarily on π-stacking interactions rather than mutation-sensitive electrostatic networks. FEP result: **ΔΔG = -0.44 kcal/mol = binding preserved** in mutant.

Biological significance

Bufadienolide's mutation-resistant binding suggests a potential **molecular basis for NOD2 signaling restoration** in R702W carriers, supporting further investigation of innate immunity. Could support **investigation of bacterial clearance** in the ~10% of Crohn's patients carrying R702W who currently lack targeted therapies.

Clinical implications

Mutation effects on drug binding are **ligand-dependent**, not universal across all compounds. Screening against wild-type protein alone would fail ~10% of patients.

CONCLUSIONS

Hypothesis supported

Deep learning combined with molecular dynamics successfully identified mutation-resistant NOD2 binders. FEP quantified predicted differential binding affinity between wild-type and R702W mutant proteins with high statistical significance (p<0.001, 8σ effect).

Key findings

- Screened 9,566 compounds → identified 2 computational lead candidates for NOD2 protein
- Febuxostat (FDA-approved for gout): binds NOD2 but **50x weaker** in R702W mutant
- Bufadienolide (natural product): **mutation-resistant** binding (ΔΔG = -0.44 kcal/mol)
- R702W causes allosteric rigidification propagating 79.4 Å from mutation to pocket
- Cysteamine negative control (0% occupancy) validates computational assay specificity

Significance

First computational drug screen targeting the NOD2 LRR domain in scientific literature. **First FEP study** quantifying how Crohn's-associated mutations affect NOD2 binding. Febuxostat identified as potential FDA-approved drug repurposing candidate for Crohn's.

Future directions

- SPR/ITC binding assays to experimentally confirm computational binding affinities
- NF-κB reporter functional assays in NOD2-expressing cell lines to test activation
- Extend FEP analysis to G908R mutation to cover ~50% of NOD2-Crohn's patients
- Screen bufadienolide structural analogs for improved drug-likeness and lower toxicity
- X-ray crystallography to validate computationally predicted binding poses and contacts

References

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 - Economou, M., et al. (2004). Differential effects of NOD2 variants on Crohn's disease risk and phenotype in diverse populations. *American Journal of Gastroenterology*, 99, 2393–2404.
- Acknowledgements:** NOD2 structure from AlphaFold DB (AF-Q9HC29). Compound libraries: COCONUT & e-Drug3D. All visualizations by finalist using PyMOL/Python.

Abstract

UT Arlington - Fort Worth Regional Science and Engineering Fair Abstract and Certification

Deep Learning Guided Discovery and FEP Validation of NOD2 Binders for the R702W Crohn's Variant

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NOD2 variants are key genetic risk factors associated with Crohn's disease, although ligand binding to NOD2 remains poorly understood. This study developed a computational pipeline to identify NOD2 LRR domain binders and measure the effect of the R702W mutation on binding affinity to NOD2. Using GNINA CNN - based docking, a library of 9,566 compounds (2,152 FDA - approved drugs, 7,414 natural products) was docked against an AlphaFold2 NOD2 model. Subsequently, using molecular fingerprints, an XGBoost classifier (NOD2 - Scout) was trained to characterize potential binders, and it was validated by scaffold split cross - validation and shuffled label control, with ROC - AUC values of 0.85 - 0.93 and approximately 0.50, respectively. Following ADMET screening, molecular dynamics simulations were performed on all 8 candidates (520 ns overall) to confirm the top candidates' stable pocket occupancy (70 - 80%) . The apo simulations indicated that the R702W mutant had reduced conformational sampling compared to the wild - type (WT) NOD2 protein (average RMSD of 3.27Å vs 4.05Å; 0% vs 12.6% of frames sampled > 5Å) . To compare binding energies between the WT and R702W mutants, Free Energy Perturbation (FEP) was implemented. FEP indicated that the R702W mutation reduces the binding affinity to NOD2 for Febuxostat (dG_bind (WT) = - 10.36 +/- 0.18kcal / mol and dG_bind (R702W) = - 8.02 +/- 0.19kcal / mol; ddG = + 2.34kcal / mol; 8 sigma, p<0.001; 50 - fold weaker in R702W) but had no significant effect on the binding affinity to NOD2 for Bufadienolide (CID 10120) (dG_bind (WT) = - 15.22 +/- 0.26kcal / mol and dG_bind (R702W) = - 15.66 +/- 0.26kcal / mol; ddG = - 0.44kcal / mol; 1.8 sigma; not significant) . No reports on the interaction of Bufadienolides with NOD2 were found. Experimental validation via SPR, ITC and HDX - MS will be needed in future studies.

Category

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